

UNIVERSITY OF LEICESTER

Time Series Forecasting Platform (Brent Crude Oil)

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Sprint 1.1 Data Collection & Research

Dataset info

- Main Target for the forecasting (Y_t) - Brent Crude Oil (collected via Yahoo Finance api)
- Exogenous Variables (X) :
  - These are the data collected to fulfil the requirement of mapping external market movers aka external factors to Brent Crude Oil price changes :
    - Global Supply - OPEC+ Total Production Volumes
      - To capture the supply targets since from the worlds largest oil cartels (aka producers) since they are the ones who cut or hikes directly the Brent pricing.
      - Data collected from EIA Open Data Portal api
      - Frequency is Monthly
    - Currency – US Dollar Index (DXY)
      - This is non-negotiable. Oil is priced in dollars; when the dollar moves, Brent moves.
      - Data Collected from Yahoo Finance api
      - Frequency is Daily
    - Market Sentiment – VIX Index (VIX)
      - This acts as a proxy for global economic fear and market sell-offs
      - Data collected from Yahoo Finance api
      - Frequency is Daily
    - Arbitrage/Competitor – WTI Crude Oil Price (WTI)
      - This gives my model the "Brent-WTI spread," which tells the model if cheaper US oil is flooding the global market.
      - Data collected from Yahoo Finance api
      - Frequency is Daily
- Since the Frequency is varying all the fluctuations will be sampled into uniform weekly frequency (Friday Close).
- The data collection period will be between 2019-2026.

(Exogenous Variables - acts as an input, affecting other variables inside the model without being influenced by them in return)

Machine Learning info

- Before building out the model there are pre-preparation to do :
  - ML Model don't inherently understand the chronological date and time
  - To tackle that the uniform weekly data frame is converted using a technique called Windowing / Lagging technique.
  - The idea is to place the past metrics side by side with future targets in a tabular grid.
  - This is the formula it will follow $Y_t = f(Y_{t-1}, X_{t-1})$

Model Category	ML Algorithm	How it affects Brent Dataset?	Project Purpose
Statistical Baseline	SARIMAX	Takes weekly Brent prices as the primary dependent variable (Y) and natively processes DXY, VIX, WTI, and OPEC data as an exogenous (X) matrix.	Parses traditional linear components to establish a hard econometric baseline.
Machine learning	XGBoost & Random Forest	Goes through chronological rows of tabular lag features (t-1, t-2) across all target and exogenous variables simultaneously.	Maps highly complex, non-linear market interactions (e.g., a massive OPEC supply drop paired with high market anxiety triggering an exponential price spike).
Future Selection	LASSO Regression	Tests linear combinations of the regressors and mathematically penalizes unhelpful variables by shrinking their coefficients to absolute zero.	Evaluates which external market drivers are genuinely influential, automatically filtering out the noise.
Deep Learning	LSTM Neural Network	Processes sequential weekly data arrays using internal "memory gates".	Preserves long-term structural supply shocks (like permanent OPEC+ quota shifts) without forgetting them as new weekly data streams in.

The regressor in the Lasso Regression is the historical data we feed to the model like the last week's oil prices, the strength of the US Dollar, and stock market fear, etc.



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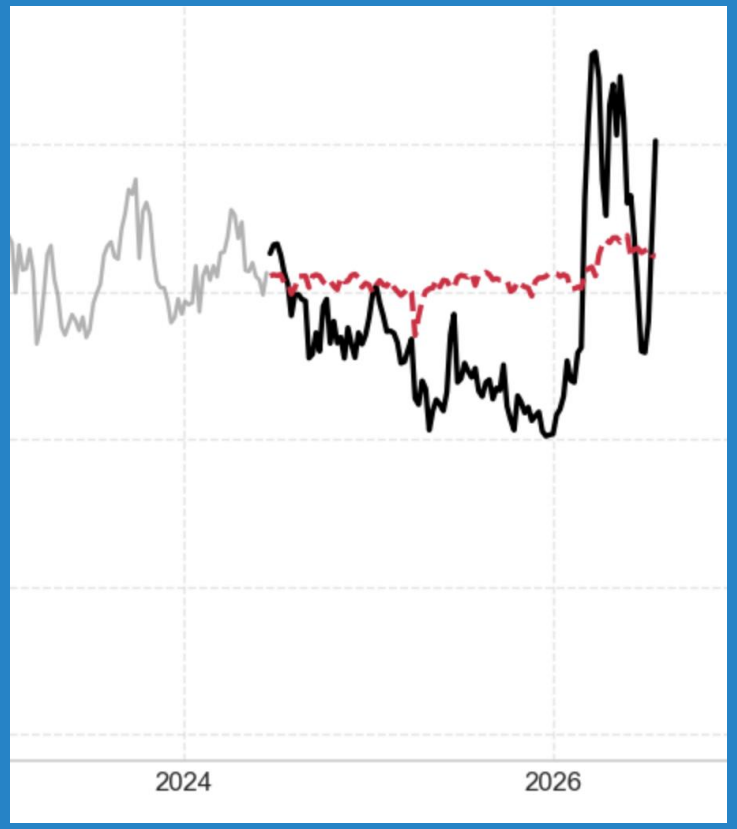
Sprint 2 - Model Building & Comparison

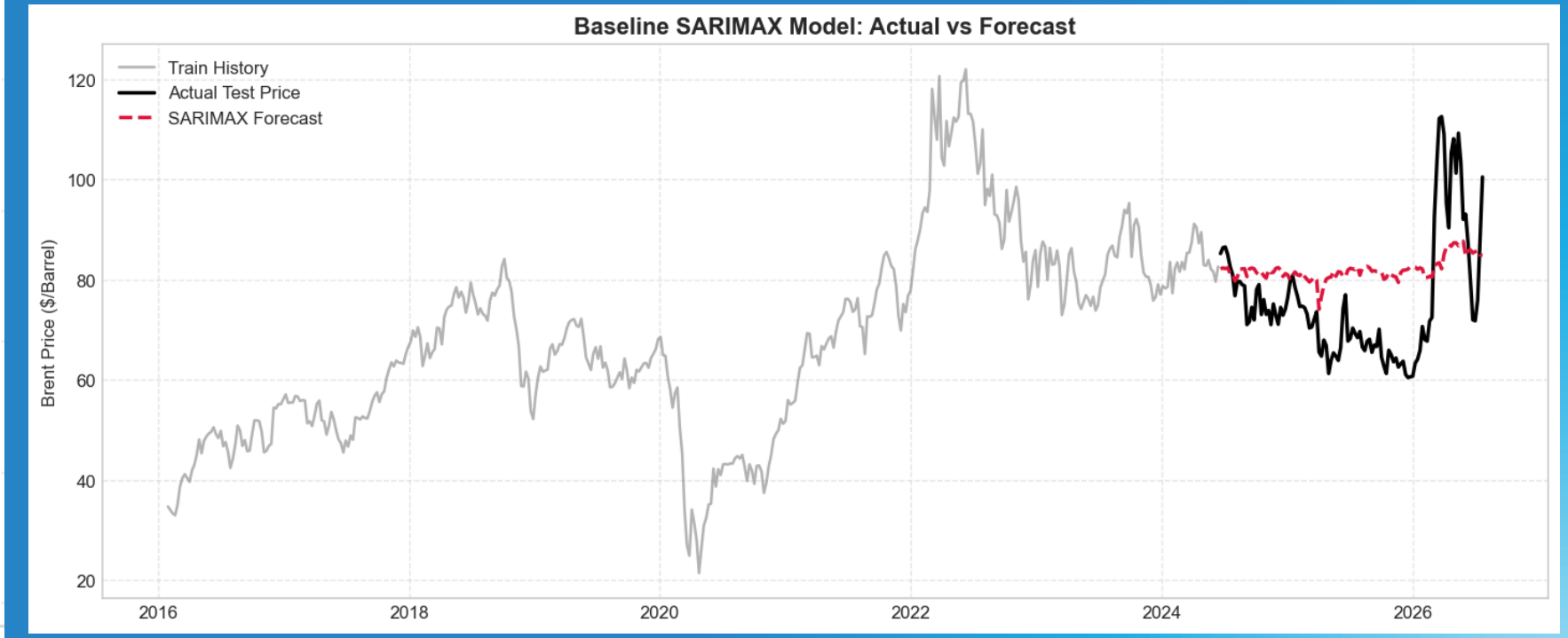
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- Establishing the Benchmark: Linear Model Limitations
- Hyperparameter Tuning & Error Reduction
- Engineering for Machine Learning: Lags & Log-Space
- Performance Comparison
- Final Best Model Selection

Establishing the Benchmark: Linear Model Limitations

- Tested an un-tuned SARIMAX(1, 1, 1) model to establish a foundational performance metric.
- **Results:** Mean Absolute Error (MAE) of $11.29 and a negative R^2 score (-0.08).
- **Conclusion:** Linear state-space models fail to capture the severe, non-linear volatility of global commodity shocks without continuous rolling updates, in other words resulting $11.29 error and negative R^2 score mathematically proved our core hypothesis: traditional linear models cannot handle the complex volatility of the oil market.

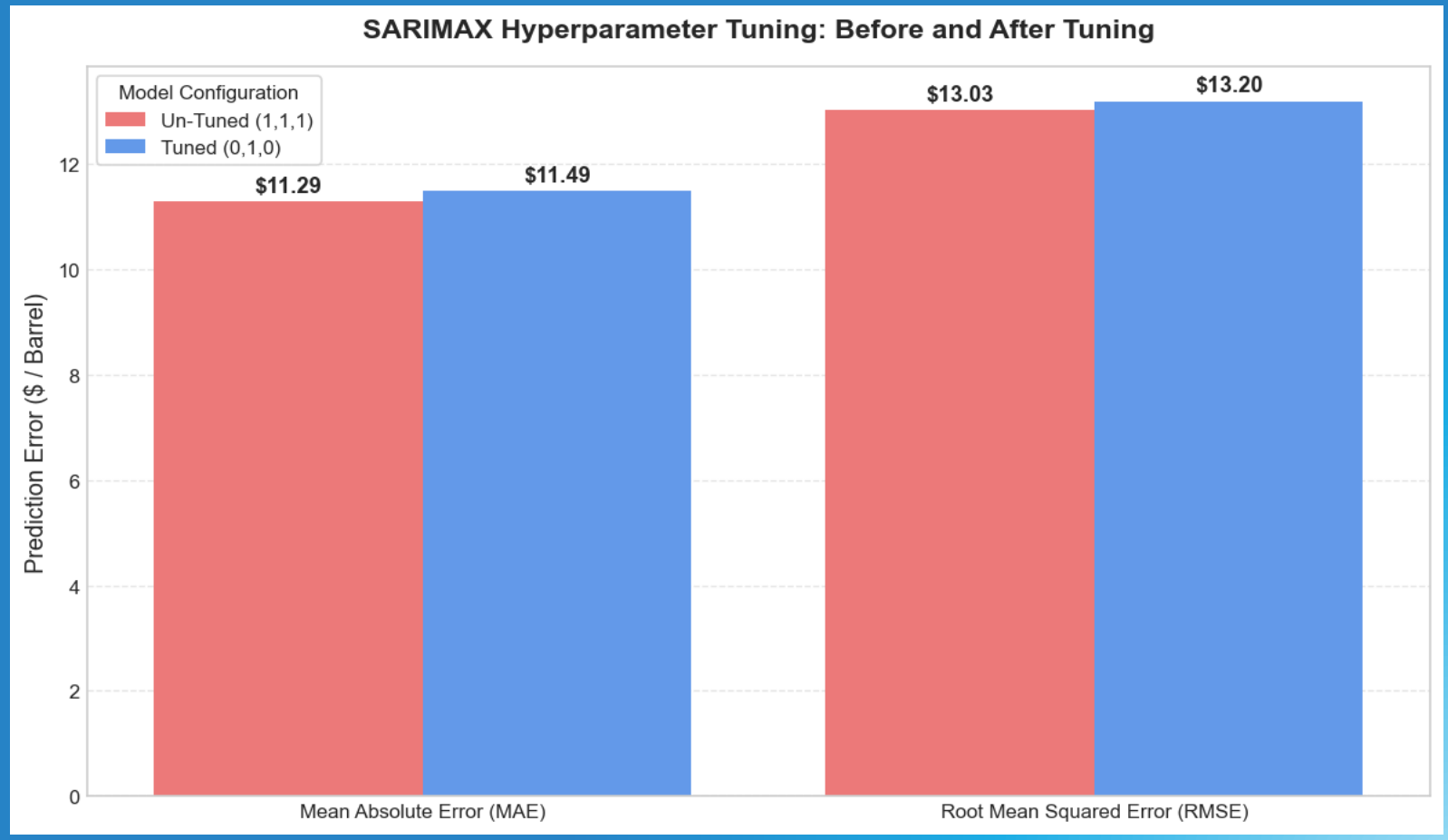




Hyperparameter Tuning & Error Reduction

- Deployed an AIC-based Grid Search to rigorously optimize the SARIMAX parameters rather than relying on assumed estimates.
- Optimal Configuration: The algorithm identified SARIMAX(0, 1, 0) as mathematically superior.
- Impact: Reduced absolute and squared prediction errors, verifying our optimization pipeline works before transitioning to advanced algorithms.

- These were the parameters I gave :
  - "p" - for how many weeks look back
  - "d" - prices changes week to week and this parameter tells me "how many times you must calculate the difference between consecutive weeks to make the overall trend flat which I set to 1 since I have known from sprint 1.2 that Brent_Crude Oil is non-stationery.
  - "q" – tells me "how many past **forecast errors** the model looks at to correct its current trajectory for its next forecast".
  - I put above parameter into the itertool lib to generate diff combination of p,d,q and then ran them into the sarimax model parameter loop for each combination generated.
  - The result being the model mathematical superior chosen config.



Engineering for Machine Learning: Lags & Log-Space

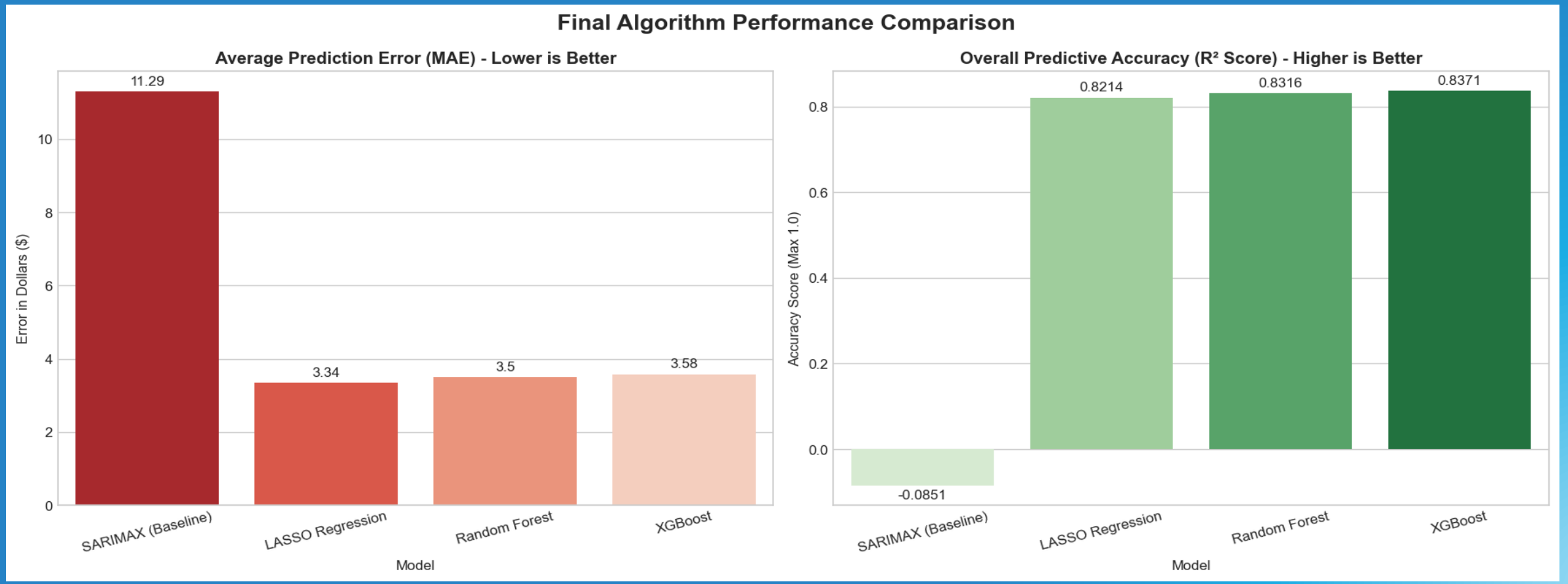
- **Data Transformation:** Converted chronological time-series data into a supervised learning matrix.
- **Lagged Features:** Engineered "short-term memory" by feeding the algorithms the exact prices from the previous 1, 2, and 3 weeks.
- **Target Normalization:** Trained all models strictly on Log-Normalized prices (under the var_name Brent_Log) to stabilize extreme variance, converting predictions back to raw US dollars (using the np.expm1) only for final evaluation.

*Calculates log exp(x) - 1 with high precision for numbers close to zero.

Performance Comparison

- **The Contenders:** SARIMAX vs LASSO Regression (Linear) ,Random Forest & XGBoost (Tree-based ML).
- **The Improvement:** All ML models slashed the $11.29 baseline error down to the ~$3.50 range.
- **Accuracy Jump:** R^2 scores increased from a failing baseline to explaining over **83%** of all market variance.

* The below Bar Chart is output visualization after doing config work on there parameters for each model.



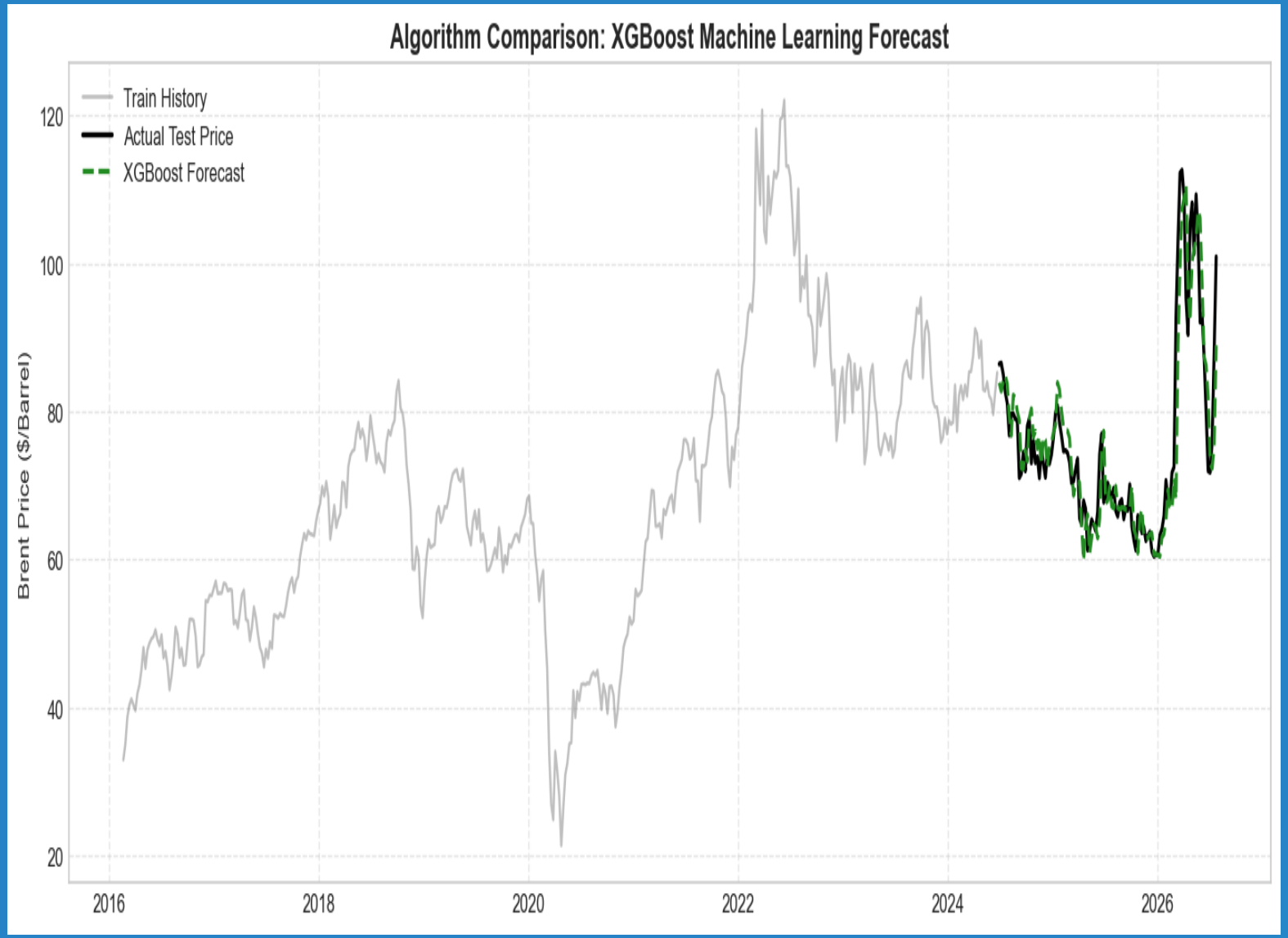
Final Best Model Selection

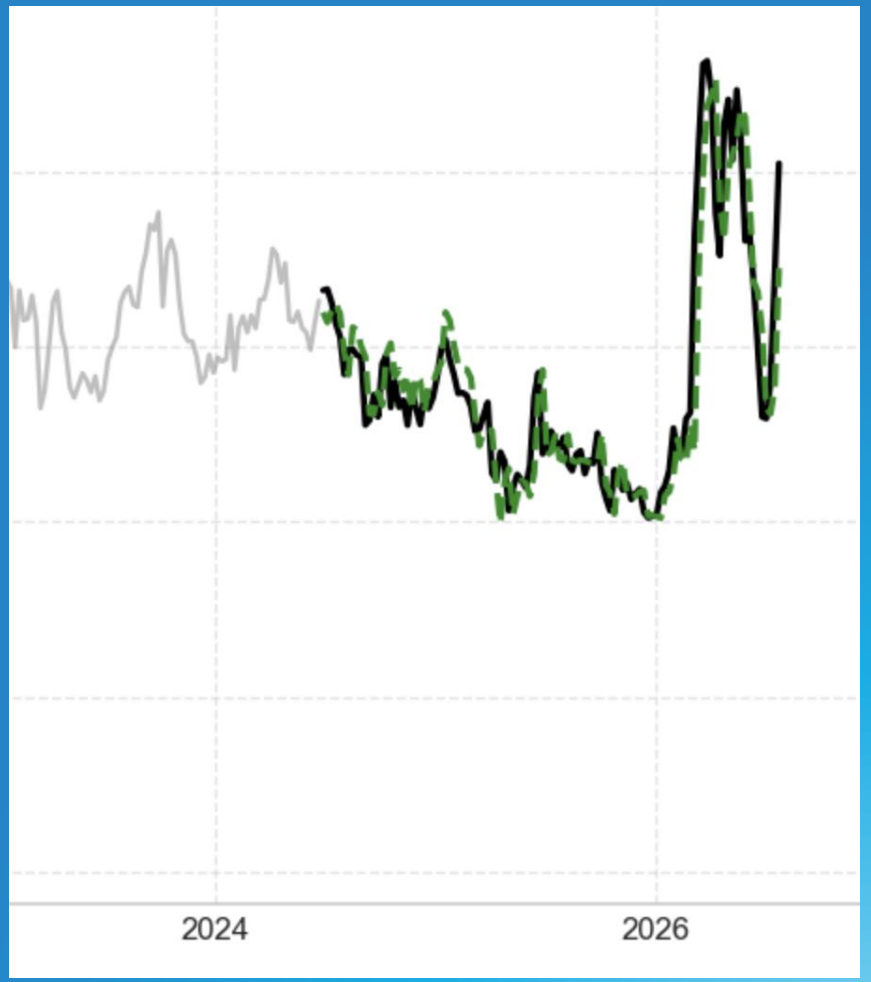
Model	MAE	RMSE	R^2
SARIMAX (Baseline)	11.29	13.03	-0.0851 = -0.8%
LASSO Regression	3.34	5.30	0.8214 = 82.14%
Random Forest	3.50	5.15	0.8316 = 83.16%
XGBoost	3.58	5.06	0.8371 = 83.71%

- **LASSO vs. XGBoost:** While LASSO achieved a slightly lower average error ($3.34), XGBoost is the superior architectural choice for volatile commodities.
- **Why XGBoost Won:** Highest overall predictive accuracy ($R^2 = 0.837$) and the lowest Root Mean Squared Error (RMSE = 5.06).
- **Strategic Advantage:** The lowest RMSE proves XGBoost is mathematically the safest and most robust model when faced with extreme, unpredictable market panics.
- In the next slide a line chart visualization will be available to further understand why XGBoost comes out on top as the best model in the selection.

* Mean Absolute Error (MAE) - tells you exactly how many dollars your model's prediction is off by, on average, every single week. So lower is always better.

* R^2 - tells you what percentage of the wild price swings (volatility) in the oil market your model successfully mapped and predicted.


Algorithm Comparison: XGBoost Machine Learning Forecast
Train History
Actual Test Price
XGBoost Forecast
Brent Price ($/Barrel)
120
100
80
60
40
20
2016
2018
2020
2022
2024
2026


2024
2026



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Sprint 3.1 - Initial Platform Building

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- Sprint 3.1 Key Accomplishments
- System Architecture & Data Pipeline changes
- Core Dashboard Modules
- Planned Enhancements

Sprint 3.1 Key Accomplishments

- Transitioned offline Jupyter notebook experiments to a modular backend (src/data_loader.py & src/models_engine.py).
- Built live market indicator cards (Brent Spot, WTI, DXY Index, VIX Volatility).
- Implemented an interactive multi-model forecast visualizer with Plotly.
- Integrated a live model evaluation matrix ($MAE, RMSE, R^2, MAPE$).
- Embedded a historical dataset explorer & summary statistics inspector.

System Architecture & Data Pipeline changes

- Data Loader (data_loader.py): Raw dataset ingestion, log transformations (Brent_Log), and automated feature engineering (lags 1–3, DXY/VIX/OPEC Z-score normalization).
- The notebook dir will now be used as the testing environment for new models to be built before deploying it onto the dashboard.
- Models Engine (models_engine.py): Unified execution across all models using a 80/20 chronological split (maintaining temporal order).
- Streamlit App (app.py): Fast frontend rendering with @st.cache_data caching.

Core Dashboard Modules

- Market KPI Metrics Bar: Live tracking of crude spot prices & daily price deltas.
- Plotly Forecast Visualizer: Interactive line chart with hover crosshairs, zooming, and toggleable model traces.
- Model Performance Matrix: Side-by-side comparative table ($MAE, RMSE, R^2, MAPE$).
- Historical Data Inspector: Expandable dataset viewer & summary metrics (describe()).

Deploy

🛢️ Brent Crude Oil Time Series Forecasting Platform (TSFP)

Brent Crude Spot Price
$90.74
↓ -6.04

WTI Crude Price
$84.46

US Dollar Index (DXY)
100.80

VIX Volatility Index
20.66

📈 Model Forecasting Results

Comparison of actual historical Brent Crude Oil prices against model forecasts (SARIMAX, LASSO, Random Forest, XGBoost, and LSTM).



📊 Model Performance Statistics

Summary evaluation statistics calculated on the test dataset split (80% Train / 20% Test).

Model	MAE ($)	RMSE ($)	R2 Score	MAPE (%)	Hyperparameters
SARIMAX (Baseline)	14.55	15.98	-0.6357	20.27	order=(1,1,1)
LASSO Regression	3.28	5.16	0.8292	4	alpha=0.01
Random Forest	3.58	5.2	0.8267	4.52	n_estimators=100, max_depth=4
XGBoost	3.52	5.17	0.8291	4.39	n_estimators=100, lr=0.1, max_depth=4
LSTM (Deep Learning)	4.93	7.73	0.6172	5.95	Units=64, Epochs=50

🔍 View Historical Market Dataset Overview

	Brent_Crude	WTI_Crude	US_Dollar_Index	VIX_Index	OPEC_Plus_Production
2026-02-20 00:00:00	71.76	66.39	97.8	19.09	34.4463
2026-02-27 00:00:00	72.48	67.02	97.61	19.86	34.4463
2026-03-06 00:00:00	92.69	90.9	98.99	29.49	27.0092
2026-03-13 00:00:00	103.14	98.71	100.36	27.19	27.0092
2026-03-20 00:00:00	112.19	98.32	99.65	26.78	27.0092
2026-03-27 00:00:00	112.57	99.64	100.15	31.05	27.0092
2026-04-03 00:00:00	109.03	111.54	100.03	23.87	25.5888
2026-04-10 00:00:00	95.2	96.57	98.65	19.23	25.5888
2026-04-17 00:00:00	90.38	83.85	98.1	17.48	25.5888
2026-04-24 00:00:00	105.33	94.4	98.51	18.71	25.5888

Summary Statistics:

	Brent_Crude	WTI_Crude	US_Dollar_Index	VIX_Index	OPEC_Plus_Production
count	552	552	552	552	552
mean	69.09	64.74	98.56	18.32	34.74
std	18.31	17.87	4.9	7.02	2.66

Planned Enhancements

- Custom CSV Upload: Interactive upload feature for custom market datasets.
- Dynamic Model Hyperparameter Tuning: Sidebar controls to adjust parameters on-the-fly.
- Automated Data Fetching: Live API integration for real-time market data ingestion.
- Adding Risk Metric / Risk-Adjusted Decision Indicator
- Need to answer to the needs of two kinds of audience :
    1. You have the professional audience where for them they will go in-depth into the data asking the question why should I buy at this price, or why should I sell now ,all the advanced thinking ? So, a visualization appropriate to them is needed.
    2. You have the unprofessional audience where to them they don't know much about the info I am giving them and just want a quick overview of what to do. Like what is the condition right now and should I buy or sell, if I buy now how much PnL would I make over a certain period, etc.
- The chart needs to be more interactive, meaning say for the model chart on the side you could have a drop-down list containing the years and for that selected year you can show the results for that particular year and when you reset it shows chart till current date.



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Sprint 4.1 Validation and Visualisation

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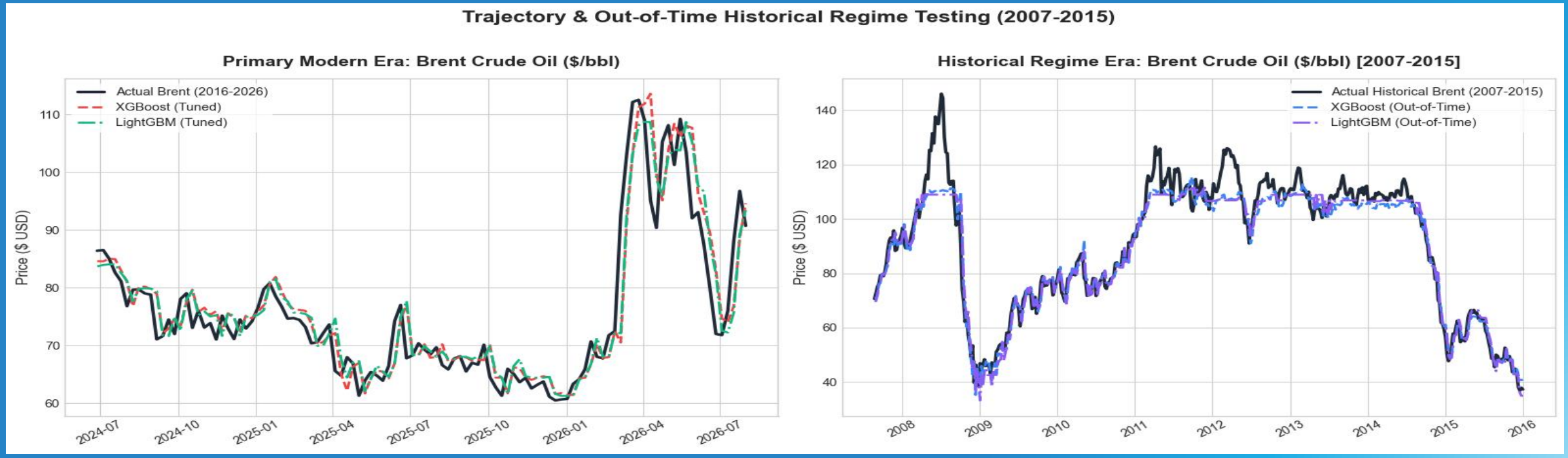
- Objectives & Methodology
- Out-of-Time Trajectory Analysis
- Cross-Era Performance Indicators
- Model Stability & Predictor Drivers

Objectives & Methodology

- Sprint Goal: Stress-test the best-performing models on an unseen secondary dataset to verify generalization and guard against modern-era overfitting.
- Secondary Dataset Setup:
  - **Era**: Historical Market Regime (2007–2015) | $N = 437$ weekly observations.
  - **Context**: Encompasses the 2008 Great Financial Crisis, capturing extreme Brent crude price swings (peaking at >$140/bbl before dropping to ~$35/bbl).
- Validated Models: Selected top-performing tree-based architecture LightGBM and XGBoost.

Out-of-Time Trajectory Analysis

- **Modern Era (2016–2026)**:
  - Both models closely follow modern price dynamics ($ MAE ~3.40$).
  - LightGBM slightly edges out XGBoost on overall error fit ($RMSE = 4.85$ vs. 5.07$).
- **Historical Crisis Era (2007–2015)**:
  - Neither model diverged during rapid structural shifts or market crashes.
  - High-volatility peaks produced temporary under-prediction, typical of non-linear tree models encountering extreme unseen price ceilings.



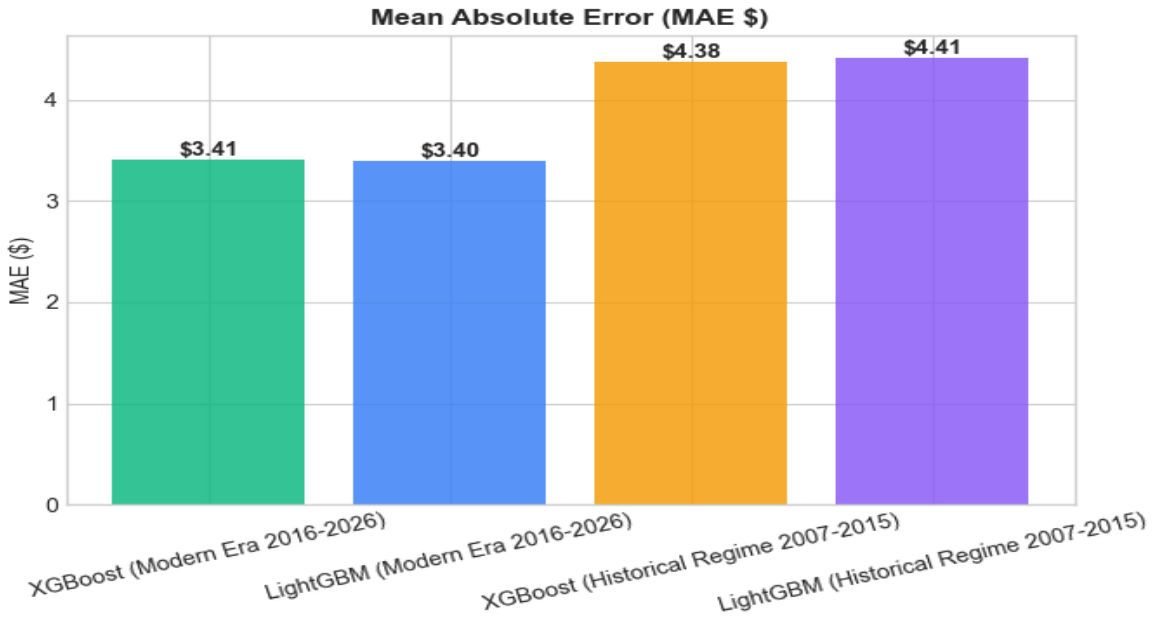
Cross-Era Performance Indicators

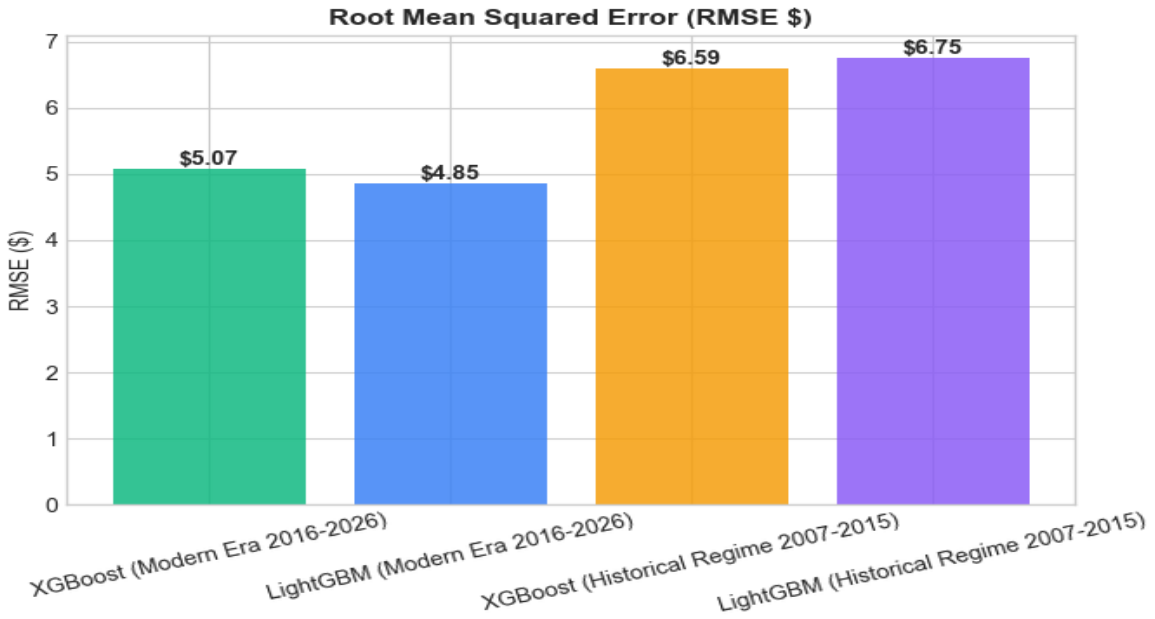
Metrics	XGBoost (Modern)	LightGBM (Modern)	XGBoost (Historical)	LightGBM (Historical)
MAE ($)	$3.41	$3.40	$4.38	$4.41
RMSE ($)	$3.41	$4.85	$6.59	$6.75
R^2 Score	0.8350	0.8491	0.9281	0.9244
MAPE (%)	4.29%	4.32%	4.70%	4.81%
Directional Acc. (%)	51.4%	44.0%	45.9%	36.5%
Durbin-Watson	1.36	1.37	0.50	0.45

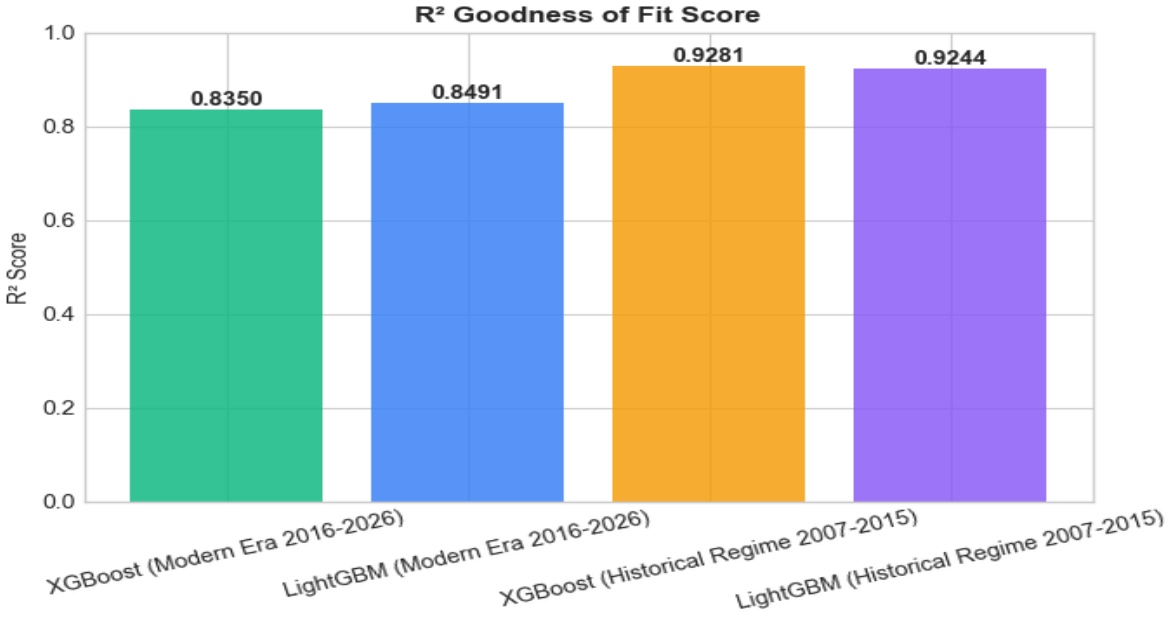
Key Take aways:

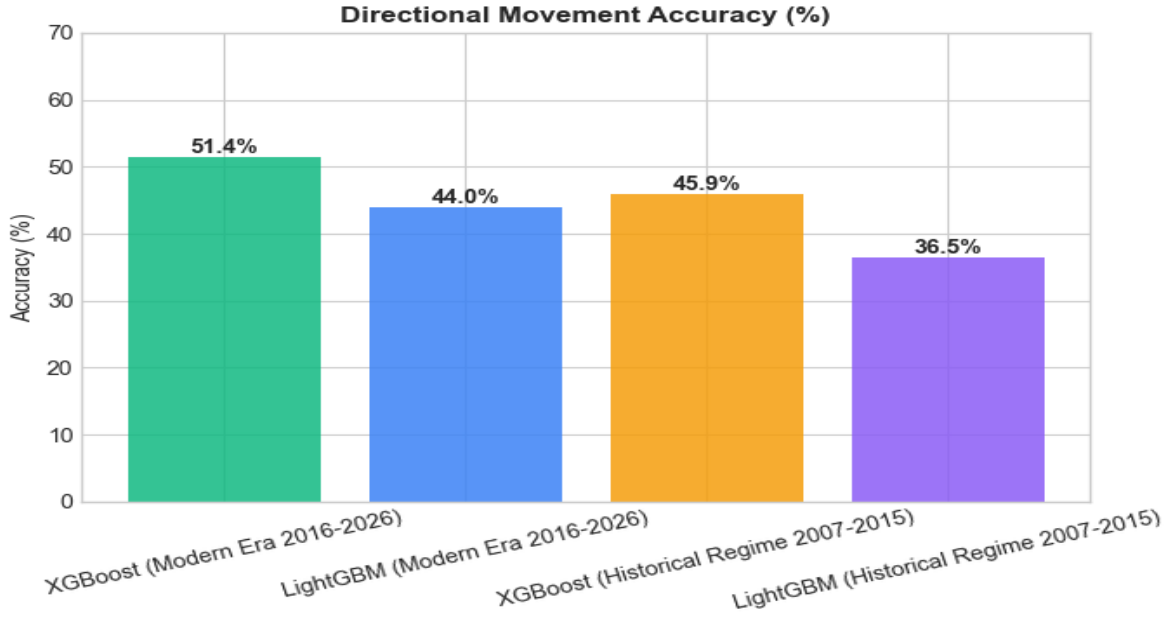
- **Strong Trend Capture**: R^2 increased to **>0.92** in the historical era, confirming excellent fit across wide swings.
- **Stable Scale Error**: MAPE remained under **4.81%** across all testing environments.
- **Absolute Error Increase**: Absolute dollar errors (MAE/RMSE) rose during 2007–2015 due to higher nominal oil prices during that era.

Performance Indicators Across Market Eras









Model Stability & Predictor Drivers

- Error Density Symmetry:
  - Residual error distributions for both eras are narrow, zero-centered bell curves, showing negligible systematic bias.
  - Historical residuals exhibit slightly broader tails due to 2008 volatility spikes.
- Feature Drivers:
  - Price_Lag_1 dominates feature importance for LightGBM, backed by macroeconomic features (VIX_Z, OPEC_Z, and USD_Z).
- Autocorrelation (Durbin-Watson):
  - Scores near ~1.37 in the modern era indicate mild positive autocorrelation, while historical scores (~ 0.45 - 0.50) reflect error persistence during crisis regimes.

Prediction Error Density Comparison Across Eras

